Codex Repository Update Instructions

Target: Mini34/pico-2w-power-transient-analyzer

Target

Update the existing private repository:

• **Repository:** `Mini34/pico-2w-power-transient-analyzer`

• **Default branch:** `main`

• **Recommended working branch:** `docs/final-ee-lab-tool-integration`

The current repository is an earlier Power/RC snapshot. This update must preserve its historical engineering evidence while bringing the repository to the final verified EE Lab Tool state.

Non-negotiable constraints

1. Do not invent or reconstruct firmware that has not been verified on the Pico. 2. Copy the exact working MicroPython files from the current ChatGPT Work/Thonny project workspace. 3. Preserve all tested GPIO assignments and hardware behavior. 4. Do not expose Wi-Fi credentials, local IP addresses, browser tabs/bookmarks, laptop branding, user names, or other personal/system identifiers. 5. Do not describe MCU temperature as ambient temperature. 6. Do not describe software fault monitoring as electrical protection or automatic cutoff. 7. Do not treat a negative fitted source resistance as valid. Flag it as invalid and retain it only as a documented test/validation issue. 8. Preserve the user's intentional behavior that temperature sampling is deferred while the device is in `STARTUP` or `MENU`. 9. Preserve the timing-sensitive calls that suppress web/temperature work during RC acquisition and diode sweep acquisition. 10. Do not change working UI, dashboard, logger, or analyzer behavior merely for stylistic cleanup.

Source bundle

Use the files in this documentation package as the authoritative repository-update input:

• `README.md`

• `docs/PROJECT_LOG.md`

• `docs/PROJECT_LOG.json`

• `docs/EE_Lab_Tool_Final_Project_Log.docx`

• `docs/EE_Lab_Tool_Final_Project_Log.pdf`

• `docs/EE_Lab_Tool_Photo_Appendix_All_26.pdf`

• `docs/EE_Lab_Tool_Design_Assets.pdf`

• `docs/PHOTO_INDEX.md`

• `images/sanitized/` (26 cropped/sanitized photographs)

• `images/designs/` (annotated overview, verified architecture, conceptual designs)

• `firmware/main.py` (the supplied current working main file)

The exact working copies of the following additional modules must be obtained from the active project workspace and committed without semantic changes unless a test proves a required correction:

- `instrument_state.py`
- `wifi_manager.py`
- `web_server.py`
- `experiment_logger.py`
- `fault_monitor.py`
- `source_characterizer.py`
- `ssd1306.py`

Exact verified hardware contract

```
| Resource | Assignment |
|---|---|
| I2C SDA | GP6 |
| I2C SCL | GP7 |
| I2C implementation | SoftI2C, 50 kHz |
| Menu/select button | GP2, active low |
| RC charge | GP0 |
| RC transistor control/discharge | GP1 |
| RC ADC | GP26 / ADC0 |
| Diode source | GP3 |
| Diode source-side ADC | GP27 / ADC1 |
| Diode forward-voltage ADC | GP28 / ADC2 |
| Internal temperature | ADC4 |
| INA219 address | 0x40 |
| OLED address | 0x3C |
```

Preserve these calibration constants in the working firmware:

```
V_SLOPE = 0.98896
V_OFFSET = -0.03531
SHUNT_OHMS = 0.1
TEMP_VREF = 3.24
```

Required repository operations

1. Create a safe branch

```
git checkout main
git pull --ff-only
git checkout -b docs/final-ee-lab-tool-integration
```

2. Preserve the original snapshot

Do not delete the original 13 diagnostic photographs or the earlier log. Move/rename the old full DOCX only if needed for clarity, for example:

```
docs/archive/EE_Summer_Project_Log_GitHub_Export_2026-08-24.docx
```

Retain Git history and avoid a destructive rewrite.

3. Replace/update documentation

- Replace the top-level `README.md` with the supplied final README.
- Replace `docs/PROJECT_LOG.md` with the supplied final Markdown log.
- Add `docs/PROJECT_LOG.json`.
- Add the final DOCX and all PDF deliverables.

- Add `docs/PHOTO_INDEX.md`.
- Add a short `docs/README.md` that links the log, photo appendix, design assets, and Codex brief.

4. Add sanitized photo assets

- Add all 26 files from `images/sanitized/`.
- Add all files from `images/designs/`.
- Do not replace these with the uncropped laptop/dashboard originals.
- Verify that no local IP such as the photographed dashboard address remains visible.
- Verify that no browser tabs, bookmarks, laptop logo/brand, or user-specific interface content remains visible.

5. Add exact working firmware

- Add the supplied `firmware/main.py`.
- Export/copy the exact working service modules listed above.
- Update `firmware/README.md` to describe deployment to the Pico.
- Add an example credentials/configuration template only if required by the working Wi-Fi manager.
- Never commit real SSIDs/passwords.

Recommended `.gitignore` additions:

```
# Local credentials and device-generated data
secrets.py
wifi_config.py
config.py
logs/
*.jsonl
__pycache__/
*.mpy
.thonny/
```

Only ignore `config.py` if the working project actually stores credentials there; do not hide a required public configuration file accidentally.

6. Source Test validation changes

Inspect the existing `source_characterizer.py` and UI workflow. Do not silently change the regression formula. Add validation around it:

- Require at least two distinct-current points; recommend at least three.
- Reject nearly identical current values.
- Confirm current polarity/sign convention before fitting.
- For the assumed source model `V = Voc - I*Rs`, require the fitted slope to be negative within a configurable tolerance.
- Calculate `Rs = -slope`.
- If `Rs < 0`, `Voc` is implausible, current span is too small, or numerical conditioning is poor, return `status = "invalid"` with a reason instead of `completed`.
- Keep the photographed `Voc = -0.003 V`, `Rs = -6.395 ohm`, `R^2 = 1.000`, 3-point result in documentation only as an invalid example.
- Add tests for a valid synthetic source and the invalid negative-Rs example.

Do not add automatic load switching or new GPIO assignments. The present mode is guided/manual.

7. Dashboard documentation and boundaries

The dashboard has already been verified. Preserve its working layout and endpoints. Documentation must state:

- MCU temperature is internal die temperature.
- Fault monitoring is monitoring/logging only.
- CSV/JSON downloads exist for experiments and faults.
- Diode PWM I-V data is display-only.
- Source Test invalid results are visibly marked.

8. Firmware validation

Perform checks appropriate for MicroPython without pretending CPython can execute hardware APIs:

- Parse/syntax-check every `.py` file.
- Search for accidental credentials and local IP addresses.
- Confirm all imports resolve on the Pico filesystem.
- Confirm no GPIO assignments changed.
- Confirm `main.py` still enters the menu and all five modes.
- Confirm RC and diode acquisition still call `service_background(allow_web=False)`.
- Confirm temperature sampling remains disabled in STARTUP/MENU per user preference.
- Confirm logging/network failures remain non-fatal.

Suggested static checks:

```
python -m compileall firmware
rg -n "password|passwd|ssid|10\.|192\.168\.|172\.(1[6-9]|2[0-9]|3[01])\." .
rg -n "GP[0-9]+|ADC[0-9]+|SoftI2C|TEMP_VREF|V_SLOPE|V_OFFSET" firmware
```

Review false positives manually; do not remove legitimate private-range examples from generic documentation unless they identify the user's actual network.

9. Link and asset checks

- Verify every README and Markdown image link resolves.
- Verify every PDF/DOCX exists and is non-empty.
- Verify the 26-photo manifest matches the 26 files.
- Verify generated/conceptual images are labeled as conceptual where applicable.
- Ensure the exact architecture diagram, not a concept render, is used as the authoritative pin-map figure.

10. Commit and pull request

Use a clear commit sequence, for example:

```
docs: add final illustrated project log and sanitized photo archive
firmware: add verified integrated EE Lab Tool sources
docs: update README for dashboard, logging, telemetry, source test, and faults
test: validate source-characterizer fit quality and invalid-result handling
```

Open a pull request against `main` with:

- a summary of all added capabilities;
- a note that the dashboard was user-verified;
- a note that source-test negative resistance remains invalid/pending refinement;
- a privacy statement confirming dashboard photos were sanitized;
- a checklist showing all 26 photographs and final PDFs were added.

Acceptance criteria

The repository update is complete only when:

- The README describes the five-mode integrated instrument, not the old two-subsystem snapshot.
- The final Markdown, JSON, DOCX, and PDF logs are present.
- All 26 sanitized photographs are present and indexed.
- The photo appendix and design-assets PDFs open correctly.
- The exact working firmware modules are present, with no credentials.
- Pin assignments and calibration constants match the verified hardware.
- Dashboard documentation covers live power, RC traces, diode results, MCU telemetry, Source Test, faults, history, and CSV/JSON export.
- Source Test invalid negative resistance is rejected/flagged.
- No automatic load bank or hardware cutoff is falsely claimed.
- All internal links and image paths pass review.
- The pull request explains what was verified on hardware and what remains future work.